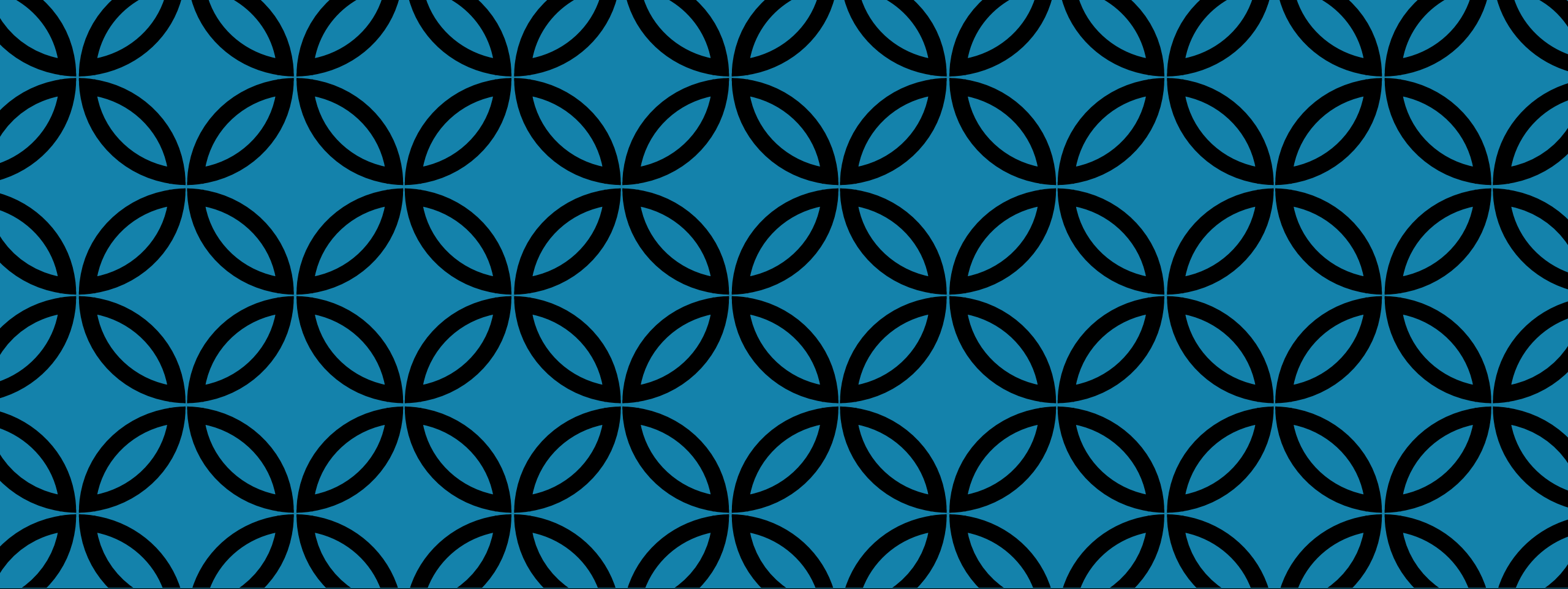




TCP-UDP

HOW MANY OF YOU USE
WHATSAPP EVERY DAY?
WHEN YOU SEND A WHATSAPP
TEXT MESSAGE, WHAT DO YOU
EXPECT?

- It should reach the receiver.
- It should not disappear.
- It should arrive correctly.
- Messages should appear in order.

EXACTLY

If I send:

Hello
How are you?
See you tomorrow.

Imagine if the receiver gets:

See you tomorrow.
Hello
How are you?

So text messages must arrive correctly and in the correct order

"NOW THINK ABOUT A GOOGLE MEET OR ZOOM VIDEO CALL."

What happens if one video frame is lost?

- The video freezes.
- The picture skips.
- It continues after a moment.

Suppose one frame of your friend's face is lost.

Should the application stop the entire video call and wait until that lost frame arrives?"

COMPARE THE TWO SITUATIONS

WhatsApp Text	Video Call
Must be accurate	Must be fast
No data loss	Small losses are acceptable
Order is important	Speed is important

So we have two different communication needs.

For text messages, **reliability is most important.**

For video calls, **speed is most important.**

NETWORK EDGE: COMMUNICATION BETWEEN DEVICES

In computer networks, communication between devices can occur in two distinct ways

1. **connection-oriented**

2. **connectionless services.**

These two types define how data is transmitted between a source and a destination

Differ mainly in their **reliability, sequencing and connection setup** mechanisms.

REGISTERED POST (TCP)

- Imagine you are sending your original degree certificate. Would you send it through ordinary mail?

Why?

- What would you choose?

- Safe
- Can track it
- Delivery confirmation

1. Registered post

2. Speed Post

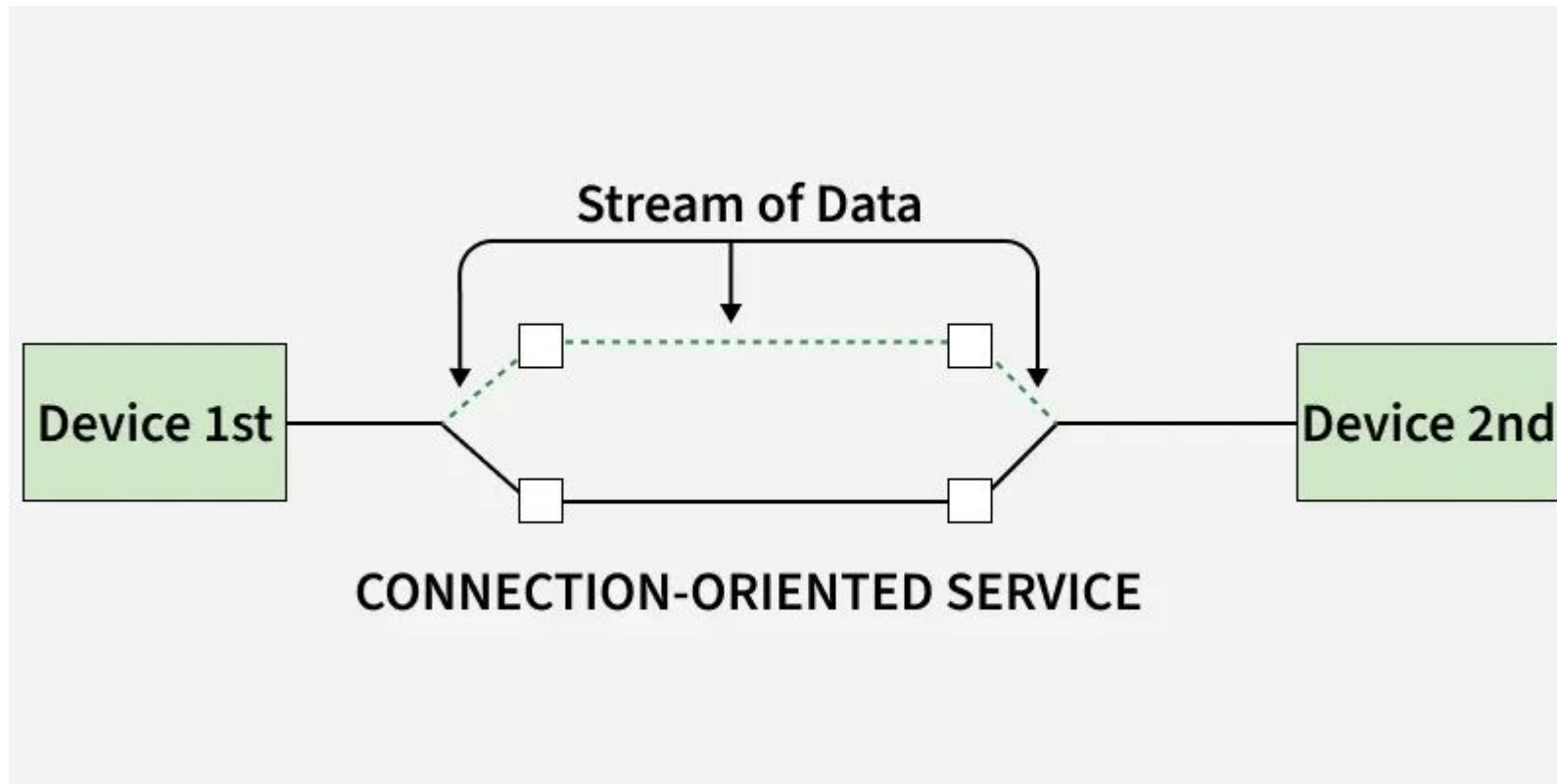
3. Courier with tracking

"This is similar to **TCP**.

TCP makes sure the data reaches safely and correctly

CONNECTION-ORIENTED SERVICE

A connection-oriented service establishes a dedicated communication path between the sender and receiver before any data is transferred. This process involves a handshake to set up the connection, ensuring both parties are ready for communication.



Once the connection is established, data packets are transmitted sequentially and reliably.

The connection remains active until all data has been successfully transferred. Afterwards, the connection is terminated.

A common example of a connection-oriented service is **TCP (Transmission Control Protocol)**, which guarantees error-free and in-order delivery of packets.

Examples:

TCP (Transmission Control Protocol): used for reliable data transfer in applications like HTTP, FTP and email.

Telephone calls: where a dedicated communication channel is established between two users.

Key Features:

Dedicated Connection: A logical/physical link is established before data transfer.

Reliable Transmission: Error detection, acknowledgement and retransmission mechanisms ensure reliability.

Sequencing: Packets arrive in the correct order.

High Overhead: Extra resources are used to establish and maintain the connection

Advantages:

Reliable and error-free data delivery

Guaranteed data sequencing

Suitable for large, continuous data transfers

Ensures retransmission of lost data

Disadvantages:

Higher latency due to connection setup

Greater resource consumption

Less efficient for small or one-time messages

TCP Handshake

by levelupcoding.co

Explained

The client sends a TCP packet with the SYN flag set and an initial sequence number to the server

1. SYN (Synchronize)



Replies with both the SYN and ACK flags set, it's an initial sequence number, and acknowledges the client's SYN packet

2. SYN-ACK (Synchronize-Acknowledge)



The client sends back an ACK packet to the server, acknowledging the server's SYN-ACK packet

3. ACK (Acknowledge)



Brought to you by

**LEVEL UP
CODING**

  @NikkiSiapno

  @ChrisStaud

NORMAL POSTCARD (UDP)

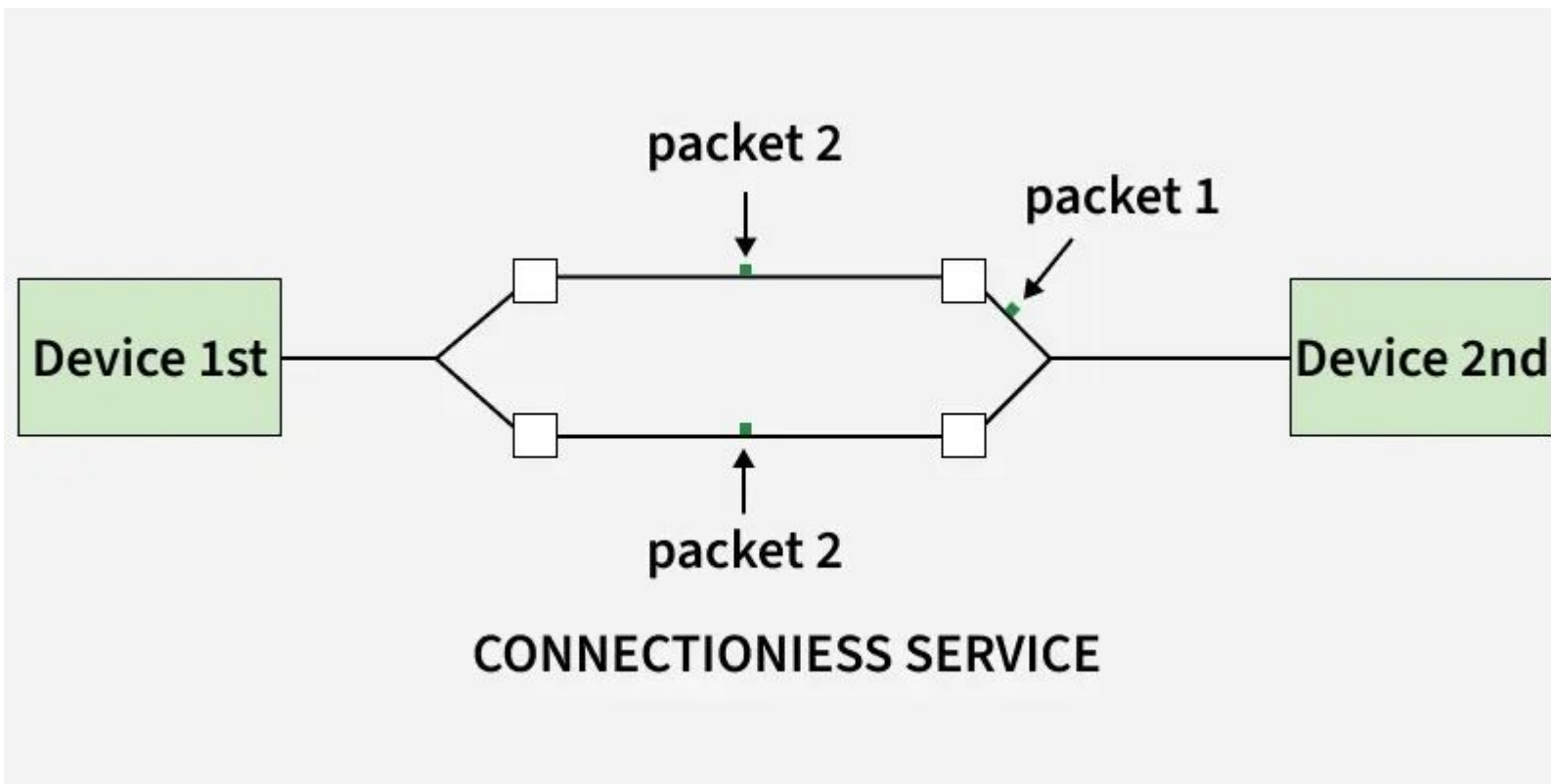
Now imagine you're sending a birthday invitation or a festival greeting. If one postcard is delayed by a day, is it a disaster?

This is similar to **UDP**.

It sends information quickly without checking whether every packet arrived.

CONNECTIONLESS SERVICE

A connectionless service transmits data packets without first establishing a dedicated path between sender and receiver. Each packet (often called a datagram) is treated independently and may follow different routes to reach the destination.



There is no guarantee of delivery order, reliability or error correction, but the transmission is faster and more scalable.

The most common example is UDP (User Datagram Protocol), used in real-time applications where speed is more important than reliability.

Examples:

UDP (User Datagram Protocol): used for DNS queries, streaming and gaming.

Postal services analogy: letters are sent independently, without confirmation of delivery.

Key Features:

No Connection Setup: Data is sent immediately.

Independent Packets: Each packet travels separately.

Faster Transmission: No overhead for connection management.

Unreliable: Packets may be lost or arrive out of order.

Advantages:

Low latency and faster communication

Efficient for small or time-sensitive data transfers

Scalable for large networks with multiple users

Simple implementation and less overhead

Disadvantages:

No error control or acknowledgment

Packets can be lost, duplicated or reordered

Not suitable for large or critical data transmissions

UDP



Use Case :- *Video Conferencing*



DIFFERENCE BETWEEN CONNECTION-ORIENTED AND CONNECTIONLESS SERVICES

Parameter	Connection-Oriented Service	Connectionless Service
Connection Setup	Requires a connection before data transfer	No connection setup needed
Reliability	Reliable; ensures delivery, error detection and retransmission	Unreliable; no delivery or error guarantee
Data Sequencing	Packets delivered in correct order	Packets may arrive out of order
Overhead	High due to connection management	Low; no connection maintenance
Speed	Slower due to setup and acknowledgments	Faster as data is sent directly
Resource Usage	Consumes more resources (buffers, control info)	Minimal resource usage
Best Suited For	File transfer, web browsing, emails (TCP)	Streaming, DNS, VoIP (UDP)
Example Protocol	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)

Every time you browse the Internet, play games, watch YouTube, or make a video call, data is travelling between your device and another computer.

But not all applications need the same type of communication.

- Some need reliability.
- Some need speed.

This is why computer networks provide two transport protocols:

TCP (Transmission Control Protocol)

UDP (User Datagram Protocol)

Real-Life Examples of TCP :

Text Communication -

We are all aware of the importance of Text Communication in the present times. Any discrepancy in texting between sender and receiver cannot be tolerated. Hence, TCP is used in Text Communication due to its reliable transmission, error control, and in-order delivery of the data.

Transfer of files or FTP -

TCP is used in File transfer when we cannot tolerate the loss of data and receiving the data in the correct order is of utmost importance. FTP uses two TCP connections i.e control connection and data connection.

Control Connection : FTP sends information like user identification and passwords.

Data Connection: In this connection, files are sent over the network.

HyperText Transfer Protocol (HTTP) -

It is used to access the data present on the World Wide Web. It uses the TCP protocol for accessing the web pages present on the internet due to the fact that TCP provides in-order data, error control, flow control, and retransmission of data segments.

Simple Mail Transfer Protocol(SMTP)

It is an application layer protocol that is used to send Emails from one system to another. SMTP uses the services of TCP to start a connection with the SMTP server. Once the SMTP server accepts the connection request, it allows the sender to send the email.

Real-Life Examples of UDP :

Online Games -

Most of the online games we play use the services of the User Datagram Protocol. Since any amount of delay cannot be tolerated in online games, UDP is widely used over TCP, which is quite slow. UDP doesn't retransmit the lost data and is a connectionless protocol, due to which it is much faster.

Ex : All online games

Video Conferencing –

Video Conferencing apps like Skype, Gmeet, Zoom, all use the services of UDP due to the fact that they are real-time applications, and any delay in receiving the data cannot be tolerated.

Ex: Skype, Google Meet, zoom, and Facetime.

Voice Over IP(VoIP) -

It is similar to Video Conferencing, where apps like Viber, WhatsApp, Google Hangouts use UDP for converting our voice to digital data and transmit it over the network, hence the name VoIP.

Ex : Viber, WhatsApp voice calling, Wi-Fi calling.

Domain Name System(DNS) -

It is a service used for mapping domain names to their corresponding IP address. The application layer uses it. It can also be viewed as a distributed database with a hierarchical name Server. DNS uses UDP for fetching the corresponding IP address due to the following reasons:

- a) UDP is much faster than TCP. After all, speed matters a lot when loading a webpage
- b) DNS requests are typically small requests and can be accommodated inside UDP segments(Header).
- c) Even though UDP is unreliable, it can be achieved in the application layer too

MEMORY TRICK

TCP

= Trustworthy

UDP

= Ultra-fast

Application

Answer

Web browsing

Online gaming

Email

Video call

DNS lookup

File download

Quick challenge

“If a packet is lost during a Zoom call, should the application wait and retransmit?”

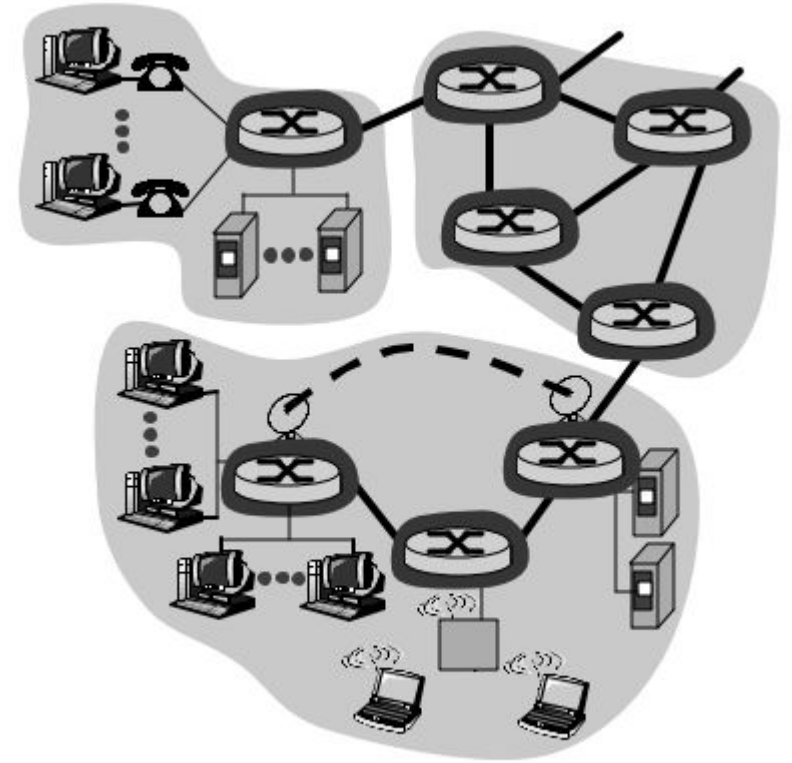
HOW IS DATA TRANSFERRED THROUGH NET?

mesh of interconnected routers

the fundamental question: how is data transferred through net?

circuit switching: dedicated circuit per call: telephone net

packet-switching: data sent thru net in discrete “chunks”



CIRCUIT SWITCHING VS PACKET SWITCHING

Imagine you want to travel from your home to college.

There are two possibilities:

1. You reserve an entire road only for yourself.
2. You use the public road along with everyone else.

Which one is more practical?

WHAT IS CIRCUIT SWITCHING?

Definition

Circuit Switching is a communication method where a **dedicated communication path** is established between the sender and receiver **before any data is transmitted**.

The path remains reserved until communication ends.

Computer A ----- Computer B

Dedicated Path

Reserved only for this communication

REAL-LIFE EXAMPLE

Telephone Call

When you make a traditional landline telephone call:

The network first connects both users.

A dedicated communication path is established.

Nobody else can use that reserved path.

After the call ends, the path is released.

Characteristics

Dedicated path

Connection setup required

Continuous communication

Predictable performance

Reliable

Advantages

✓ No congestion

✓ Guaranteed bandwidth

✓ Constant delay

✓ Suitable for voice communication

Disadvantages

✗ Resources remain reserved even when nobody is speaking.

Example:

Imagine two people are silent for 30 seconds during a phone call. The network still reserves the entire communication path. This wastes bandwidth.

WHAT IS PACKET SWITCHING?

Definition

Packet Switching divides data into **small packets**.

Each packet travels independently through the network.

No dedicated path is reserved.

Each packet may travel through a different route.

Large File



Packet 1

Packet 2

Packet 3

Packet 4



Router



Destination

REAL-LIFE EXAMPLE

Courier Service

Suppose you are sending 100 books.

Instead of one huge truck,
the courier company sends several small vehicles.

Some may reach earlier.

Some may take another road.

Finally,

all books arrive at the destination.

CHARACTERISTICS

No dedicated path

Shared bandwidth

Efficient resource utilization

Packets may follow different routes

Possible congestion



Advantages

- ✓ Efficient
- ✓ Supports many users
- ✓ Better bandwidth utilization
- ✓ Cost-effective
- ✓ Ideal for the Internet

Disadvantages

- ✗ Delay
- ✗ Congestion
- ✗ Packet loss
- ✗ Packets may arrive out of order

WHY DOES THE INTERNET USE PACKET SWITCHING?

Suppose one million people are using the Internet. Can everyone have a dedicated communication path ?

NO

It would require enormous resources.

Instead,

the Internet shares the available bandwidth among all users.

This is Packet Switching

COMPARISON TABLE

Feature	Circuit Switching	Packet Switching
Path	Dedicated	Shared
Connection Setup	Required	Not required
Resource Usage	Reserved	Shared
Speed	Constant	Depends on congestion
Delay	Predictable	Variable
Cost	High	Low
Example	Telephone Network	Internet

EASY MEMORY TRICK

Circuit Switching

"Reserve First → Communicate Later"

Packet Switching

"Break → Send → Reassemble"